



Geopolitical Event-Driven Portfolio Risk Analysis Platform

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INTRODUCTION

- **Background**

The Cboe Volatility Index (VIX), often called the "fear gauge," indicates how turbulent the market is expected to be; higher numbers signal fear and increased risk, while lower numbers suggest complacency.

- **Key Questions**

Markets react to world events such as wars, sanctions or coups. But how?

- **Solution**

Our platform explains the geopolitical drivers behind market fear

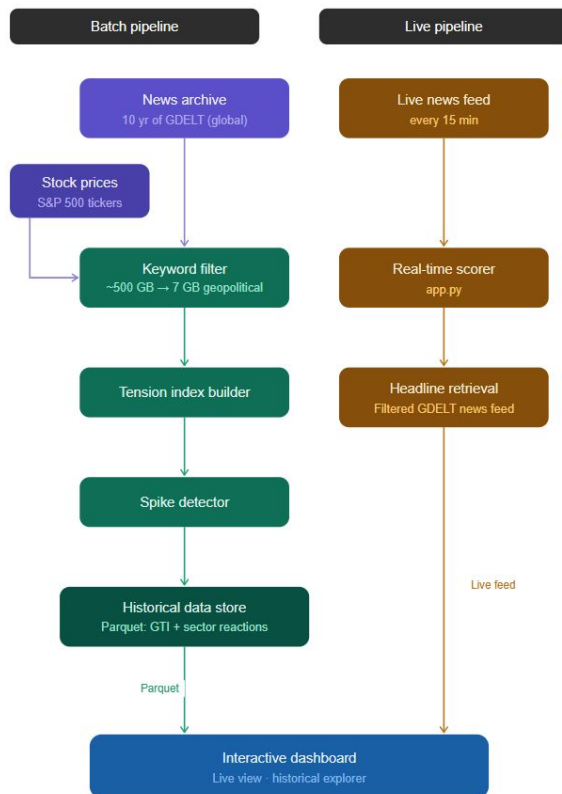
WHY BIG DATA?

- GDELT - GLoBal Database of Events, Language, and Tone
 - Monitors news media worldwide, in real time, across dozens of languages - Updates every 15 minutes.
- One month of GDELT data is roughly ~26GB. With a decade of data, this exceeds ~2TB.
- Joined with S&P 500 price data: 500 tickers $\times$ 31 spikes $\times$ ± 30 trading days ~1M rows
- Far too much data to be stored locally

WHY PYSPARK?

- Keyword filtering across billions of records cannot run on a laptop
- ± 30 day joins across 500 tickers would cause out-of-memory errors in pandas
- Daily Geo-Tension Index requires groupBy aggregation across 7.6M articles
- Parquet columnar storage integrates natively — dashboard loads in milliseconds

SYSTEM ARCHITECTURE



DATA PIPELINE - 1

- News Sentiment -> Tension Index -> Anomaly Detection -> Market Impact -> Visualization.
- We wrote a bash script that downloads GDELT files and filters for geopolitical keywords — WAR, SANCTION, MILITARY, TERROR, PROTEST, WEAPON, NUCLEAR — simultaneously.
- We tested four approaches (1 month of data) :
 - Download only: 15 min, 26 GB
 - Download then filter (sequential): 44 min, 58 MB
 - Download + filter simultaneously (final): 5 min, 58 MB(Dataset spans more than 10 years — simultaneous filtering was the only viable approach)

DATA PIPELINE - 2

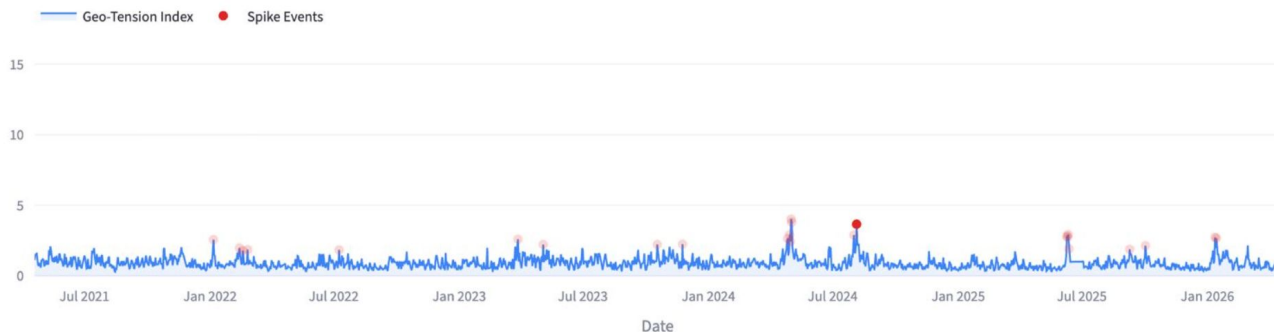
- Geo-Tension Index
 - PySpark reads the 5GB filtered TSV from HDFS, filters to international articles only, and computes our index daily.
- Spike detection and market reaction
 - Detect tension spikes
 - Pull S&P 500 price data
- All aggregated results are fed to our live dashboard

SPIKE DETECTION

- Spike events are detected using a z-score approach

$$\text{Spike} = \text{geo_tension_index} > \text{yearly_mean} + 3 \times \text{yearly_std}$$

- Per-year threshold
 - Compute mean and standard deviation per year, not globally.
- Why 3-sigma?
 - 3 -sigma gave the clearest separation between everyday news cycles and genuine geopolitical shocks



THE GEO-TENSION INDEX

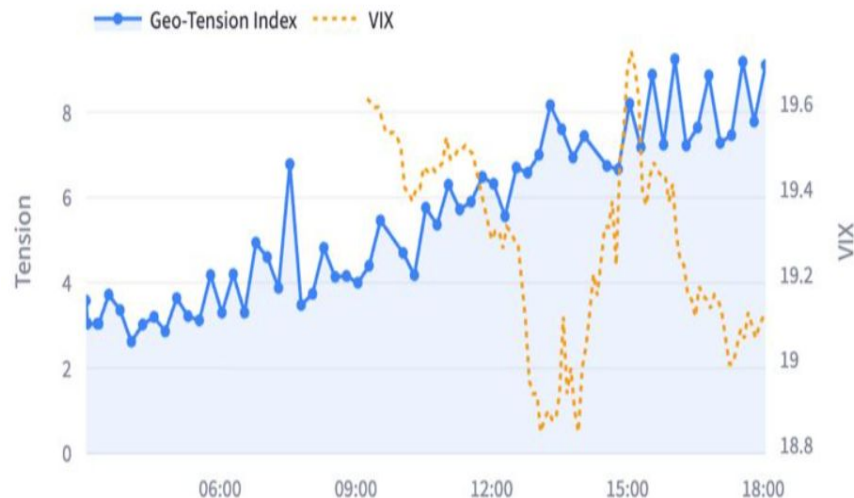
- We extract 7/60 columns per GDELT record
 - record_id, date, source, url, themes, tone, and category
- Tone is the key signal
 - a string of sentiment scores computed by GDELT for each article
- Extract and aggregate that pre-computed negative score daily.
- Themes column to filter for international articles

$$geo_tension_raw = avg(negative_tone) \times log(article_count + 1)$$

- Logarithmic scaling weights higher-volume events without letting volume dominate

VIX vs GEO-TENSION INDEX

- VIX reflects expected market volatility. Our index reflects geopolitical news tension
- Case 1: Both spike
 - A major geopolitical shock dominates market sentiment
- Case 2: VIX spikes
 - Financial stress without geopolitical cause
- Case 3: Geo-Tension spikes
 - Geopolitical tension rises, but markets do not react strongly



DASHBOARD DEMO

VALIDATION & FINDINGS

- Geo-Tension Index leads VIX during geopolitical events — news moves first, markets follow
- Sector reaction patterns are consistent across 31 spike events, suggesting a repeatable market signal
 - Energy and Defense consistently outperform
 - Technology and Airlines consistently underperform
 - Gold behaves as a safe haven
 - Russia-Ukraine (Feb 2022): Energy +5.8%, Defense +3.2%, Tech -2.1%

“WHY NOT CHECK THE NEWS?”

- Our index precedes VIX
 - news moves first, markets follow.
By the time VIX spikes, markets may have already reacted
- A single article cannot tell you market impact. We aggregate millions to quantify tension level
- We provide 10 years of historical context backing our decisions

ENGINEERING CHALLENGES

CHALLENGE	SOLUTION
2TB+ raw data	Simultaneous download and filter process
Data quality issues	Multi-layer regex filtering
Rate limits	Batch collection and pre-computed Parquet
Weekend trading gaps	Nearest trading day fallback
Spark datetime bug with ancient dates	Date range filter and CORRECTED rebase mode.

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THANKS!

Q&A
